

Ans-1

If all actors are equally popular, each actor has the probability $P_i = \frac{1}{6}$

Total $N = 150$

∴ Expected frequency for each actor

$$E_i = N \times \frac{1}{6} = 150 \times \frac{1}{6} = 25$$

Hypotheses:

H_0 : All actors equally popular

H_A : Not all actors equally popular

$$df = K - 1 = 6 - 1 = 5$$

$$\alpha = 0.05$$

$$\therefore \text{Critical value} \left|_{5, 0.05} \Rightarrow \chi^2_{0.05, 5} = 11.070\right.$$

Now,

$$\chi^2 = \sum_{i=1}^6 \frac{(O_i - E_i)^2}{E_i}$$

∴ Actor A, $O = 24, E = 25$

$$(O - E) = -1, (O - E)^2 = 1, \frac{1}{25} = 0.04$$

Actor B, $O = 20, E = 25$

$$(O - E) = -5, (O - E)^2 = 25, \frac{25}{25} = 1$$

Actor C, $O = 32, E = 25$

$$(O - E) = 7, (O - E)^2 = 49, \frac{49}{25} = 1.96$$

Actor D, $O = 25, E = 25$

$$(O - E) = 0, (O - E)^2 = 0, \frac{0}{25} = 0$$

Actor E, $O = 28$, $E = 25$

$(O-E) = 3$, $(O-E)^2 = 9$, $\frac{9}{25} = 0.36$

Actor F, $O = 21$, $E = 25$

$(O-E) = -4$, $(O-E)^2 = 16$, $\frac{16}{25} = 0.64$

$\therefore \sum \chi^2 = 0.04 + 1 + 0 + 1.96 + 0.36 + 0.64 = 4$

$\therefore \chi^2_{calc} = 4$

$\therefore$ Comparing χ^2_{crit} and χ^2_{calc} , $4 < 11.070$

$\therefore$ At 5% Significance level, there is no statistically significant evidence that the six actors differ in popularity.

$\therefore H_0$ is failed to be rejected.

$$\frac{(28-25)^2}{25} = \frac{9}{25} = 0.36$$

$$\frac{(21-25)^2}{25} = \frac{16}{25} = 0.64$$

$$\frac{(25-25)^2}{25} = \frac{0}{25} = 0$$

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ANS-2

Cells (two observations per cell) :

$$A=1, B=1 : 65, 70 \rightarrow T_{11} = 135, \bar{y}_{11} = 67.5$$

$$A=1, B=2 : 20, 40 \rightarrow T_{12} = 60, \bar{y}_{12} = 30$$

$$A=2, B=1 : 30, 35 \rightarrow T_{21} = 65, \bar{y}_{21} = 32.5$$

$$A=2, B=2 : 50, 40 \rightarrow T_{22} = 90, \bar{y}_{22} = 45$$

Totals :

Total number of observations, $N = 8$

$$\text{Grand total, } G = \sum y = 350$$

$$\text{Grand mean, } \bar{y} = 350/8 = 43.75$$

Number of levels : $a=2, b=2$, replications per cell $r=2$

Marginal totals and mean:

$$\text{Operator } A=1, T_{A1} = 135 + 60 = 195, \bar{y}_{A1} = \frac{195}{4} = 48.75$$

$$\text{Operator } A=2, T_{A2} = 65 + 90 = 155, \bar{y}_{A2} = \frac{155}{4} = 38.75$$

$$\text{Machine } B=1, T_{B1} = 135 + 65 = 200, \bar{y}_{B1} = \frac{200}{4} = 50$$

$$\text{Machine } B=2, T_{B2} = 60 + 90 = 150, \bar{y}_{B2} = \frac{150}{4} = 37.50$$

Now,

$$\begin{aligned} \sum y^2 &= 65^2 + 70^2 + 20^2 + 40^2 + 30^2 + 35^2 + 50^2 + 40^2 \\ &= 17350 \end{aligned}$$

$$\frac{G^2}{N} = \frac{350^2}{8} = \frac{122500}{8} = 15312.5$$

Total SS,

$$SS_T = 17350 - 15312.5 = 2037.5$$

$$\begin{aligned} SS_A &= \frac{(195)^2}{4} + \frac{(155)^2}{4} - 15312.5 \\ &= 9506.25 + 6006.25 - 15312.5 \\ &= 200 \end{aligned}$$

$$\begin{aligned} SS_B &= \frac{(200)^2}{4} + \frac{(150)^2}{4} - 15312.5 \\ &= 10000 + 5625 - 15312.5 \\ &= 312.5 \end{aligned}$$

$$\begin{aligned} SS_{\text{cells}} &= \sum \frac{T_{ij}^2}{n} - \frac{G^2}{N} \\ &= \frac{135^2}{2} + \frac{60^2}{2} + \frac{65^2}{2} + \frac{90^2}{2} - 15312.5 \\ &= 9112.5 + 1800 + 2112.5 + 4050 - 15312.5 \\ &= 1762.5 \end{aligned}$$

$$\begin{aligned} SS_{AB} &= SS_{\text{cells}} - SS_A - SS_B = 1762.5 - 200 - 312.5 \\ &= 1250 \end{aligned}$$

$$SS_E = SS_T - SS_{\text{cells}} = 2037.5 - 1762.5 = 275$$

$$df_T = 8 - 1 = 7$$

$$df_A = 1, \quad df_B = 1, \quad df_{AB} = 1, \quad df_E = 4$$

Mean Squares:

$$MS_A = \frac{200}{1} = 200$$

$$MS_B = \frac{312.5}{1} = 312.5$$

$$MS_{AB} = \frac{1250}{1} = 1250$$

$$MS_E = \frac{275}{4} = 68.75$$

F calculation:

$$F_A = \frac{200}{68.75} = 2.9091$$

$$F_B = \frac{312.5}{68.75} = 4.5455$$

$$F_{AB} = \frac{1250}{68.75} = 18.1818$$

ANOVA TABLE:

SOURCE OF VARIATION	SS	df	MS	F
Operator A	200	1	200	2.9091
Machine B	312.5	1	312.5	4.5455
A x B	1250.0	1	1250	18.1818
Error	275	4	68.75	—
Total	2037.5	7	—	—

Critical value:

for $\alpha = 0.05$, $df_1 = 1$, $df_2 = 9$

$$F_{0.05; 1, 9} = 7.71$$

$$\rightarrow F_{AB} = 18.18 > 7.71 \quad (\text{Reject } H_0)$$

$$\rightarrow F_A = 2.91 < 7.71 \quad (\text{Accept } H_0)$$

$$\rightarrow F_B = 4.55 < 7.71 \quad (\text{Accept } H_0)$$

Because the interaction is significant, the main effects (A and B) should be interpreted with caution $\rightarrow$ the effect of machine depends on the operator (and vice-versa).

	2M	7L	22	Source of Variation
Operator A	500	1	500	
Machine B	215.2	1	215.2	
A x B	1520	1	1520	
Total	2175.2		2175.2	
Error			2175.2 - 1520 - 500 - 215.2 = 940	
Total			2175.2 + 940 = 3115.2	

Ans-3

$$\text{Grand total } G = \sum y = 176.5$$

$$N = 16$$

$$\text{Grand mean, } \bar{y} = \frac{176.5}{16} = 11.03125$$

$$\text{Row totals (drivers), } T_R = [44, 43, 44, 45.5]$$

$$\text{Column totals (cars), } T_C = [49, 45, 44, 43.5]$$

$$\text{Treatment (blend) totals,}$$

$$T_A = 53, \bar{y}_A = \frac{53}{4} = 13.25$$

$$T_B = 48, \bar{y}_B = 12$$

$$T_C = 40.5, \bar{y}_C = 10.125$$

$$T_D = 35, \bar{y}_D = 8.75$$

let,

$$n = 4, N = n^2$$

degrees of freedom,

$$df_R = n - 1 = 3$$

$$df_C = 3$$

$$df_{\text{treat}} = 3$$

$$df_E = (n-1)(n-2) = 6$$

$$df_T = N - 1 = 15$$

Now,

$$\sum y^2 = 1996.25$$

(correction term,

$$C = \frac{G^2}{N} = \frac{176.5^2}{16} = \frac{31152.25}{16}$$

$$= 1947.015625$$

Total SS,

$$SS_T = 1996.25 - 1947.015625 = 49.234375$$

$$\sum T_R^2 = 44^2 + 43^2 + 44^2 + 45.5^2 = 7791.25$$

$$SS_R = \frac{7791.25}{4} - C = 1947.8125 - 1947.015625 = 0.796875$$

$$\sum T_C^2 = 44^2 + 45^2 + 44^2 + 43.5^2 = 7789.25$$

$$SS_C = \frac{7789.25}{4} - C = 1947.3125 - 1947.015625 = 0.296875$$

$$\sum T_{\text{treat}}^2 = 53^2 + 48^2 + 40.5^2 + 35^2 = 7978.25$$

$$SS_{\text{treat}} = \frac{7978.25}{4} - C = 1994.5625 - 1947.015625 = 47.546875$$

$$SS_E = SS_T - (SS_R + SS_C + SS_{\text{treat}})$$

$$= 49.234375 - (0.796875 + 0.296875 + 47.546875)$$

$$= 0.59375$$

$$MS_R = \frac{6.796875}{3} = 0.2265625$$

$$MS_C = \frac{0.296875}{3} = 0.09895833$$

$$MS_{\text{Treat}} = \frac{47.546875}{3} = 15.84895833$$

$$MS_E = \frac{0.59375}{6} = 0.09895833$$

F-value,

$$F_R = \frac{MS_R}{MS_E} = \frac{0.2265625}{0.09895833} = 2.289$$

$$F_C = \frac{MS_C}{MS_E} = \frac{0.09895833}{0.09895833} = 1$$

$$F_{\text{Treat}} = \frac{MS_{\text{Treat}}}{MS_E} = \frac{15.84896}{0.09895833} = 160.13$$

ANOVA TABLE

SOURCE	SS	df	MS	F
Rows (drivers)	0.7969	3	0.2656	2.689
Columns (cars)	0.2969	3	0.0990	1
Treatments	47.5469	3	15.8490	160.13
Error	0.5938	6	0.09896	—
Total	49.2344	15	—	—

Conclusion:

$$\rightarrow F_{\text{treat}} = 160.13 > 4.76 \rightarrow \text{Reject } H_0$$

$$\rightarrow F_R = 2.084 < 4.76 \rightarrow \text{Accept } H_0$$

$$\rightarrow F_c = 1 < 4.76 \rightarrow \text{Accept } H_0$$

Because the Latin-Square design has removed row and column variation, The very large treatment MS (relative to tiny error MS) indicates that differences among blends are real and highly significant.

	1	2	3	4
1	2235.0	2170.0	2140.0	2110.0
2	2170.0	2235.0	2110.0	2140.0
3	2140.0	2110.0	2235.0	2170.0
4	2110.0	2140.0	2170.0	2235.0

Ans-4

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$\text{Where, } \binom{n}{x} = \frac{n!}{x!(n-x)!}$$

Given,

$$p = 0.02$$

$$n = 50$$

$$x = 3$$

$$\therefore \binom{n}{x} = \frac{50 \times 49 \times 48}{3 \times 2 \times 1} = \frac{117600}{6} = 19600$$

$$p^3 = (0.02)^3 = 0.000008$$

$$(1-p)^{47} = (0.98)^{47} \approx 0.385$$

$$\begin{aligned} P(X=3) &= 19600 \times (0.000008) \times 0.385 \\ &= 19600 \times 0.00000308 \\ &\approx 0.0603 \end{aligned}$$

$\therefore$ the probability of finding exactly 3 defective items in a random sample of 50 is approximately 6%.

Ans-5

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

Where,

$\lambda = 5$ (mean number of arrivals per hour)

$k = 7$ (probability of exactly 7 arrivals)

$$5^7 = 78125$$

$$7! = 5040$$

$$e^{-5} \approx 0.006737$$

$$P(X=7) = \frac{e^{-5} \cdot 5^7}{7!} = \frac{0.006737 \times 78125}{5040}$$

$$= \frac{526.17}{5040}$$

$$\approx 0.1044$$

So, there is about a 10.4% chance that exactly 7 customers will arrive in the next hour.

ANS-6

We have :

Number of patients ; $n = 100$

Probability of side effects ; $p = 0.01$

To get, $P(X > 2)$; where X = number of patients who experience side effects.

$\therefore$ n is large and p is very small.

$\therefore$ Expected value, $np = 100 \times 0.01 = 1$; is small

So, Here, poisson approximation to the binomial distribution.

Justification:

A Binomial $B(n, p)$ is well approximated by

Poisson(λ) when

$\rightarrow n$ is large

$\rightarrow p$ is small

$\rightarrow \lambda = np$ is moderate (< 5)

Here, $\lambda = np = 100 \times 0.01 = 1$

So, we use,

$X \sim \text{Poisson}(\lambda=1)$

Now,

$$P(X=0),$$

$$P(0) = e^{-1} \frac{1^0}{0!} = e^{-1} = 0.3679$$

$$P(X=1),$$

$$P(1) = e^{-1} \frac{1^1}{1!} = e^{-1} = 0.3679$$

$$P(X=2),$$

$$P(2) = e^{-1} \frac{1^2}{2!} = e^{-1} \frac{1}{2} = 0.1839$$

$$\begin{aligned}\therefore P(X \leq 2) &= 0.3679 + 0.3679 + 0.1839 \\ &= 0.9197\end{aligned}$$

$$\therefore P(X > 2) = 1 - 0.9197 = 0.0803$$

$\therefore$ There is about 8% chance that more than 2 patients will experience side effects.

ANS-7

Given,

Claimed population mean (μ_0) = 100 hours

Sample size (n) = 25

Sample mean ($\bar{x}$) = 98 hours

Sample standard deviation (s) = 5 hours

Significance level (α) = 0.05

Hypotheses:

$$H_0 : \mu = 100$$

$$H_1 : \mu \neq 100$$

This is a two-tailed test

for a one-sample t -test,

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$\text{Standard error (SE)} = \frac{s}{\sqrt{n}} = \frac{5}{\sqrt{25}} = 1$$

$$t = \frac{98 - 100}{1} = -2$$

degrees of freedom,

$$df = n - 1 = 25 - 1 = 24$$

at, $\alpha = 0.05$,

$$t_{\text{crit}}(0.025, 24) \approx \pm 2.064$$

Now,

Reject H_0 if:

$$t < -2.064 \quad \text{or} \quad t > 2.064$$

but, $t = -2$

$$\therefore -2.064 < -2 < 2.064$$

$\therefore$ The test statistic does not fall in the rejection region.

$\therefore$ Thus, we fail to reject the null Hypotheses (H_0)

Conclusion:

At the 5% significance level, there is not enough evidence to conclude that the mean lifetime of the batteries differs from 100 hours

The sample mean (98 hours) is slightly lower, but not statistically different from the company's claim.